

Problem-Solution Fit

Date	15 March 2026
Team ID	XXXXXXX
Project Name	OPTICROP – SMART AGRICULTURAL PRODUCTION OPTIMIZATION ENGINE
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] ☐ Farmers ☐ Agricultural Researchers ☐ Agriculture Officers ☐ Agribusiness Consultants ☐ New Farmers	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] ☐ Limited technical knowledge ☐ Limited internet access in rural areas ☐ Lack of awareness about soil analysis ☐ Budget constraints	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Traditional Farming Experience • Pros: Easy to follow • Cons: Less accurate and not data-driven Agricultural Experts • Pros: Professional guidance • Cons: Limited availability and higher cost Online Farming Resources • Pros: Easily accessible
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		Cons: General recommendations, not personalized
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Difficulty in selecting suitable crops (High) - Low crop yield due to poor crop selection (High) - Lack of data-driven farming decisions (Medium) - Dependence on traditional farming methods (High)	9. PROBLEM ROOT / CAUSE [RC] ☐ Lack of scientific crop selection methods ☐ Insufficient knowledge about soil nutrients ☐ Variability in weather conditions ☐ Limited access to intelligent decision-support systems	7. BEHAVIOR + ITS INTENSITY [BE] - Frequently seeks farming advice (High) - Regularly checks weather conditions (Medium) - Uses traditional methods for crop selection (High) - Interested in technology-assisted farming (Medium)
3. TRIGGERS TO ACT [TR] ☐ Crop failures ☐ Low agricultural productivity ☐ Financial losses ☐ Need for better farming decisions ☐ Changing weather conditions	10. YOUR SOLUTION [SL] OPTICROP – SMART AGRICULTURAL PRODUCTION OPTIMIZATION ENGINE - Uses Machine Learning to recommend suitable crops - Analyzes soil nutrients and weather conditions - Provides accurate crop predictions - Supports data-driven farming decisions - Improves productivity and profitability - Reduces risk of crop failure	8. CHANNELS OF BEHAVIOR [CH] ONLINE - Agricultural websites - Mobile applications - Government agriculture portals - Social media farming groups OFFLINE - Agriculture officers - Farmer meetings - Local farming communities - Agricultural training programs
4. EMOTIONS BEFORE / AFTER [EM] Before: - Confused - Worried - Uncertain - Frustrated After: - Confident - Satisfied		

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| <ul style="list-style-type: none">• Optimistic• Motivated | | |
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